



Regularized Feature Selection Landscapes: An Empirical Study of Multimodality

Xavier F. C. Sánchez-Díaz^(✉) , Corentin Masson ,
and Ole Jakob Mengshoel 

Norwegian University of Science and Technology, Trondheim, Norway
{xavier.sanchezdz,ole.j.mengshoel}@ntnu.no, camasson@stud.ntnu.no

Abstract. The processing of features in data is among the key topics in machine learning. While a broad range of heuristics for feature processing, including feature selection, have been developed and experimented with, less research has been concerned with the underlying fitness landscape. In this paper, we perform a fitness landscape analysis of feature selection, using local optima networks and other methods. We focus on the impact of regularization, an important element of many machine learning methods. Our study using ten datasets and learning of decision trees confirms and adds to previous findings that feature selection landscapes are highly multimodal. It is the first study to focus on the impact of regularization on the landscape induced by feature selection. In the ten datasets studied, we find a high degree of multimodality when there is no regularization. With increasing regularization, the degree of multimodality generally drops off but remains substantial.

Keywords: Machine Learning · Feature Selection · Regularization · Landscape Analysis · Multimodality · Local Optima Networks · Visualization

1 Introduction

Context. Feature selection is a machine learning (ML) problem that has been getting more important as datasets continue to grow, both in terms of number of features and number of cases [15, 24, 28]. Redundant and irrelevant features not only reduce the performance of supervised learning models, but also force users to gather more data than required, potentially increasing operating costs while decreasing interpretability [15, 48].

In feature selection, a distinction is often made between filter, wrapper, and embedded methods [24]. Variants of local search (such as backward selection and forward selection) have traditionally been employed for wrapper-based feature selection [15]. More recently, other methods including genetic algorithms [18, 19, 26], stochastic local search [33, 63], memetic algorithms [50], item sets [23], regression [22], and deep learning [31, 58, 63] have been used.

The plethora of methods studied may suggest that the feature selection problem is not only a complex one, perhaps in part due to the multimodality of its

fitness landscape, but also a problem of broad interest to the scientific community. Feature selection problems seem to be multimodal [34], even if this has not been emphasized in the literature, with some exceptions [6, 50, 59].

Motivating Challenges. There is a broad and deep literature on multimodal optimization [29, 43]. However, the problem of feature selection is under-researched and the following observations can be made [35]. *First*, much research on multimodal optimization has focused on synthetic or continuous optimization problems [21, 36, 45]. Feature selection, on the other hand, is a combinatorial, real-world optimization problem. *Second*, in cases where evolutionary algorithms have been used for combinatorial problems, including feature selection [18, 52, 61], there has been surprisingly little emphasis on maintaining diversity using niching [60], although there are some exceptions [6, 59]. At the same time, diversity maintenance has been argued to be beneficial when optimizing multimodal problems [29, 43], and there are indications that feature selection induces multimodal fitness landscapes [34]. *Third*, multimodal optimization algorithms often search for “as many local optima as possible” without explicit guidance [25, 51]. Machine learning including feature selection may be computationally demanding, and thus it may be better to compute the desired number of local optima directly rather than computing too many local optima and then delegating the pruning task to be carried out manually. However, the degree to which the “as many local optima as possible”-method is problematic for feature selection is not clear, and depends on the nature of the fitness landscape, which again depends on the dataset, the ML algorithm, and the degree of regularization.

Contributions. In this paper we study the feature selection problem from an empirical fitness landscape perspective, with particular emphasis on better understanding how multimodality varies for different regularization settings. Our contributions include the following:

- We establish the number of local optima for different datasets when using decision trees for machine learning. Our ten relatively small datasets enable an exhaustive study of the corresponding fitness landscapes. We establish that all ten datasets induce highly multimodal fitness landscapes in which the number of local optima ranges from 17 up to 50,985.
- We visualize and analyze how the local optima vary under different regularization settings. We find that the number of local optima drops with increasing degree of regularization, but the drops vary dramatically across different datasets and degree of regularization. For the highest level of regularization that we study, the number of local optima ranges from 6 to 133.
- We make datasets and source code publicly available, thus enabling reproducibility and future research.

Overall, we seek to improve the understanding of the multimodal nature of feature selection problem in this paper. We firmly establish the multimodal nature of ten different datasets via our empirical study of decision tree learning. The hope is that this empirical study will inform future research on multimodal optimization methods, including evolutionary niching methods, for feature selection and other combinatorial problems.

2 Background

2.1 Feature Selection for Machine Learning

Feature selection has been utilized in many domains, including for text classification and categorization [13, 62], for visual recognition and scientific data [7], and for biological and medical data sets [22]. While deep learning, including deep neural networks, has alleviated some of the challenges of feature engineering including feature selections, challenges remain. These challenges include explainability, interpretability, and understandability [39]. These things are especially important in high-stakes decision making that impacts human health and safety [3, 27, 48]. For high-stakes decision making, model-based interpretability, in which feature selection is crucial, can be argued to be essential [39, 48].

Decision trees are typically interpretable and used for classification or regression [4, 46]. We focus in this paper on classification trees. In this case, the leaf nodes contain class labels while the remaining nodes represent features. Intuitively, a decision tree is structured such that the most distinguishing features are closest to the root. The quality of a feature is computed while learning a tree. The better a feature is in distinguishing between different class labels in the examples of the dataset, the more important the feature is, and the closer it is to the root of the tree. Decision trees have also been used incrementally [57], in ensembles known as random forests [5], and in hybrids with other AI or ML methods [27, 64].

To some extent, decision tree learning performs embedded feature selection [44, 53] by not including irrelevant and redundant features in the tree. Unfortunately, the method is still prone to overfitting by growing too large decision trees, in other words including too many features [4, 47]. Clearly, feature selection using a filter or wrapper is a general approach to counter-act this problem,¹ an approach that also works with ML models beyond decision trees.

Despite much progress on heuristic optimization algorithms, the landscapes of many real-world fitness functions are not well understood [29, 43]. This is certainly the case for the fitness landscapes of feature selection [34]. In this paper, we are (i) using an existing ML method, decision tree learning [4, 46, 47], to exhaustively create fitness function samples and (ii) producing results that improve the understanding of feature selection fitness landscapes, especially regarding the number and structure of local and global optima, thereby enabling next-generation multimodal optimization methods.

2.2 Landscape Analysis and LONs

Many benchmarking fitness functions—including the Ackley, Griewank, Rastrigin, Schwefel Sine, Shekel’s Foxhole, Sine Envelope, Eggholder, and Weierstrass functions—are multimodal and have been extensively studied in the literature [21, 36]. However, many of these test functions are synthetic or continuous. Thus,

¹ Other methods, not studied here, include decision tree pruning [4, 47].

there is the question of how they compare to fitness functions found in combinatorial applications. Although exploratory landscape analysis started for continuous landscapes [21, 36], such landscape analysis has also been conducted for the combinatorial domain [17, 40, 41, 55] including feature selection [37, 38].

Especially relevant is the use of suitable visualization methods for multimodal combinatorial domains, where the concepts of ruggedness and neighborhoods are difficult to visualize [1, 49]. For this, we explore Local Optima Networks (LONs) [12, 42]. A LON is a compact representation of the fitness landscape in the form of a weighted graph, and it shows the size and direction of the basins of attraction [11] across the landscape. We augment LONs with other visualization methods as discussed in Sect. 3 and Sect. 4.

2.3 Feature Selection, Regularization, and Multimodality

We specifically focus on the interaction between feature selection and regularization. Regularization is important in many different areas of machine learning [8, 15, 16, 56, 61, 65], as it holds relevance even for recent machine learning methods such as deep convolutional neural networks (CNNs) [9]: “A critical factor in training concerns the network’s regularization, which prevents the structure from overfitting. This work analyzes several regularization methods developed in the past few years, showing significant improvements for different CNN models.”

Feature selection can be understood as a pseudo-Boolean function optimization problem [15, 33, 34]. Much existing research has developed significant results by focusing on pseudo-Boolean functions with rather few local optima. In contrast, highly multimodal pseudo-Boolean functions remain a challenge [25, 29]. This raises these questions: How multimodal are feature selection problems? How does multimodality change as the degree of regularization is varied? Existing research highlights feature redundancy, multimodality, and distribution of local optima in the feature subsets [32, 37, 38]. Building on these findings, we return to our answers to these questions—which are based on empirical studies of feature selection fitness landscapes with varying degrees of regularization—in Sect. 4.

3 Methods and Models

Similar to previous research on continuous synthetic benchmark fitness functions [36, 45], we focus on landscapes rather than optimization methods. We perform an empirical study with decision tree learning on several feature selection problems. As a consequence, we get a global picture of the effect of varying degree of regularization, as discussed in Sect. 3.2 about experimentation.

3.1 Definitions and Methods

In this paper, we explore fitness landscapes [1] of the form $(\mathcal{X}, f, \mathcal{N})$, where $\mathcal{X}$ is the search space, f is a fitness function to optimize, and $\mathcal{N}$ is a neighborhood relation of some kind.

We consider feature selection as a pseudo-Boolean function (PBF) optimization problem. The search space $\mathcal{X}$ is the set of bitstrings $\mathbb{B}^n$ of length n . We optimize (without loss of generality, maximize) a function of the form $f : \mathbb{B}^n \rightarrow \mathbb{R}_{\geq 0}$:

$$\mathbf{b}^* = \arg \max_{\mathbf{b} \in \mathbb{B}^n} f(\mathbf{b}). \quad (1)$$

As feature selection landscapes are multimodal, we are interested in a set of multiple local optima. A solution $\mathbf{b}^*$ is considered a local optimum (in this paper, a minimum) if it is not larger than all its neighbors:

$$\forall \mathbf{b}' \in N(\mathbf{b}^*), f(\mathbf{b}^*) \leq f(\mathbf{b}'). \quad (2)$$

We define a set of local optima $\mathbf{B}_L^* = \{\mathbf{b}_1^*, \mathbf{b}_2^*, \dots\}$ where each $\mathbf{b}_i^*$ is a local optimum. We differentiate between the set of local optima $\mathbf{B}_L^*$ and the set of global optima $\mathbf{B}_G^*$, or their shorthand $\mathbf{L}$ and $\mathbf{G}$ respectively. An optimum $\mathbf{b}_i^*$ is said to be global when $f(\mathbf{b}_i^*) > f(\mathbf{b}_j^*)$ for all $\mathbf{b}_j^* \in \mathbf{L}$, given that $\mathbf{b}_j^* \neq \mathbf{b}_i^*$.

For feature selection, we interpret the presence of a 1 in a bitstring $\mathbf{b}$ as using that feature for training a classifier, while a 0 means we do not use it. We model the feature selection problem as an *energy* function to minimize:

$$h(\mathbf{b}) = h_E(T(\mathbf{b})) + \epsilon \cdot h_P(\mathbf{b}), \quad (3)$$

where $h_E(T(\mathbf{b}))$ is the classification error over a given dataset, using an ML model T (in this paper, a decision tree). Regularization (or penalty) is formalized by $h_P(\mathbf{b})$, which penalizes according to the number of features (or 1-bits) used for training with the feature subset $\mathbf{b}$. This type of feature-penalization is common in ML [8, 15, 61]. The ϵ parameter controls the degree of regularization: by using small values for ϵ , we put more weight on minimizing the classification error. On the other hand, larger values for ϵ , encourage using as few features as possible.

Since we deal with bitstrings, the neighborhood relation $N : \mathcal{X} \rightarrow 2^{\mathcal{X}}$ assigns *neighbors* to every solution in the search space $\mathcal{X}$. In this paper, we use a subset of this neighborhood relation more akin to the standard definition of local neighborhood $\mathcal{N}(\mathbf{b})$: a set of states reachable in one bit flip (or all states with a Hamming distance of 1) from a given state $\mathbf{b}$.

Given any hill-climbing algorithm HC , e.g., the 1+1 EA [10], the *basin of attraction* of a local optimum $\mathbf{b}^*$ is the set $\{\mathbf{b} \in \mathbb{B}^n \mid HC(\mathbf{b}) = \mathbf{b}^*\}$. Basins of attraction are the foundations of LONs. A LON is a weighted graph of the form $G = (V, E)$, where the set of vertices V is the set of local optima, and E is the set of basin-transition edges. Given two different local optima $\mathbf{b}_1^*$ and $\mathbf{b}_2^*$, an edge $e_{12} \in E$ links their corresponding vertices if there are neighboring points in their respective basins of attraction. The weight of this edge is the number of such pairs, $w(e_{12}) = |\{(\mathbf{b}_i, \mathbf{b}_j) \in \mathbb{B}^{2n} \mid HC(\mathbf{b}_i) = \mathbf{b}_1^*, HC(\mathbf{b}_j) = \mathbf{b}_2^*, \mathbf{b}_i \in \mathcal{N}(\mathbf{b}_j)\}|$.

3.2 Experimental Settings and Data

We study 10 datasets from the University of California, Irvine (UCI) ML Repository [20]. We consider relatively small- n classification datasets that can be

exhaustively enumerated. The datasets are presented in Table 1, and the computed accuracy tables are available in the paper’s repository [54].

On each dataset, we train a decision tree and compute an accuracy table that comprises all possible 2^n subsets of features. The decision tree implementation is provided by the BetaML.jl package in Julia [30], and we use the default hyperparameters² with a 70/30 train/test split.

While there is little research reported on hyperparameter (HP) tuning for decision tree learning in the literature, we focus here on two relevant studies by Gomes Mantovani et al. [14] and Aouedi et al. [2]. Gomes Mantovani et al. [14] empirically study the impact of HP parameter tuning using 94 datasets. They observe that “[T]uning techniques can significantly improve the predictive performance of the [decision trees]. Depending on the dataset, tuning techniques’ performance values can be very small compared to those obtained by default HP settings. Hence, the results indicate that it is better to use the default settings for some optimization problems.” They also find differences between CART [4] and C4.5 [47], in that HP tuning has a greater positive impact on CART. Aouedi et al. report several experimental findings about different tree models for ML, including the decision tree and random forest models [2]. In particular, they investigate the impact of varying the HPs `max_depth` and `min_samples_split`. Their main finding is that “the difference between tuned and default [hyperparameters] of [random forest] and [decision tree] is small,” and report for decision trees that “accuracy ranges from 81.51% with its default parameters to 82.31% with the tuned version.” These experimental results support our use of BetaML.jl’s default HPs for decision tree learning.

4 Experimental Results

4.1 Experiment 1: Impact of Varying Regularization

Goal. The goal of this experiment is to examine the modalities of the ten datasets summarized to the left of Table 1 and see if and how they change when introducing different values of the regularization parameter ϵ .

Design. As discussed in Sect. 3.2, for each dataset we construct accuracy tables via exhaustive enumeration of all $\mathbf{b} \in \mathbb{B}^n$ feature subsets. After the decision tree has been trained on 70% of the dataset using features indicated by $\mathbf{b}$, it is tested on the remaining 30%. Then, and depending on the model’s accuracy, we calculate its energy using Eq. 3. For the regularization term, we test with $\epsilon \in \{0, 1/32, 1/16, 1/8\}$.

Results and Discussion. To the right in Table 1 we present the number of local optima (including global optima) for each dataset, under different regularization

² We used these hyperparameter values for the `DecisionTreeClassifier` of BetaML: `max_depth=0` (no limits), `min_gain=0.0`, `min_records=2`, `max_features=0` (consider all features), `splitting_criterion=BetaML.Utils.gini` (Gini impurity index [4]), and `rng = Random.GLOBAL_RNG` (no set seed for random number generation).

Table 1. Datasets used in this study, sorted by number of features (n). We present the number of examples m , as well as the number of local optima $|\mathbf{L}|$ and global optima $|\mathbf{G}|$ for various values of the regularization term ϵ . We highlight in bold the two datasets that we analyze further in Sect. 4.2.

Name	Subject Area	n	m	Number of optima							
				$\epsilon = 0$		$\epsilon = 1/32$		$\epsilon = 1/16$		$\epsilon = 1/8$	
				$ \mathbf{L} $	$ \mathbf{G} $	$ \mathbf{L} $	$ \mathbf{G} $	$ \mathbf{L} $	$ \mathbf{G} $	$ \mathbf{L} $	$ \mathbf{G} $
1-seeds	Biology	7	210	20	5	14	1	6	1	7	1
2-e-coli	Biology	7	336	17	1	7	1	7	1	6	1
3-breast-w	Health and Medicine	9	699	65	2	6	1	9	1	9	1
4-glass	Physics and Chemistry	9	214	65	1	51	2	22	2	7	2
5-heart-c	Health and Medicine	13	303	700	1	407	1	117	1	13	1
6-wine	Physics and Chemistry	13	178	976	58	286	2	29	3	14	3
7-credit-a	Business	15	690	2511	4	351	1	18	1	15	1
8-zoo	Biology	16	101	10862	8275	3003	1	134	1	16	1
9-letter-r	Computer Science	16	20000	3964	1	16	1	16	1	16	1
10-hepatitis	Health and Medicine	19	155	50985	2284	23577	3	7621	3	133	3

values ϵ . The first thing to notice is the nontrivial number of optima $|\mathbf{L}|$ for all datasets. Even for the two smallest datasets with 7-bit long bitstrings, we have $|\mathbf{L}| = 20$ and $|\mathbf{L}| = 17$ local optima respectively. At the other end of the spectrum we have more than 50,000 optima for hepatitis. Secondly, it is of interest to highlight how the number of optima varies when the regularization parameter ϵ increases. As we increase the fitness penalty on the number of features in the fitness function h , we indirectly reshape the landscape since the model prefers bitstrings with fewer bits sets to 1. Although the expected behavior is a decrease in the number of optima, the decrease is quite significant but highly variable between datasets. See for example the difference for the hepatitis dataset in the last row, for $\epsilon = 1/8$ with $|\mathbf{L}| = 133$ versus $\epsilon = 0$ with $|\mathbf{L}| = 50985$.

4.2 Experiment 2: A Study of Regularization in Two Datasets

Goal. A steep reduction in the number of optima suggests that the landscapes undergo several changes due to regularization. In this experiment, we aim to describe such changes by studying two datasets in detail.

Design. We take a closer look at two datasets—4-glass and 5-heart-c—with $|\mathbf{L}| = 65$ and $|\mathbf{L}| = 700$ local optima respectively in the unregularized case. We visualize their optima comparing the landscapes under regularization values of $\epsilon = 0$ and $\epsilon = 1/8$. Additionally, we observe the three lowest-energy optima $\mathbf{b}_i^*$ to look for redundant features, and how they change under regularization.

Results and Discussion. Figures 1 and 2 show a general overview of the 4-glass and 5-heart-c datasets, respectively. In these 2D bitmaps, we *slice* the bitstring

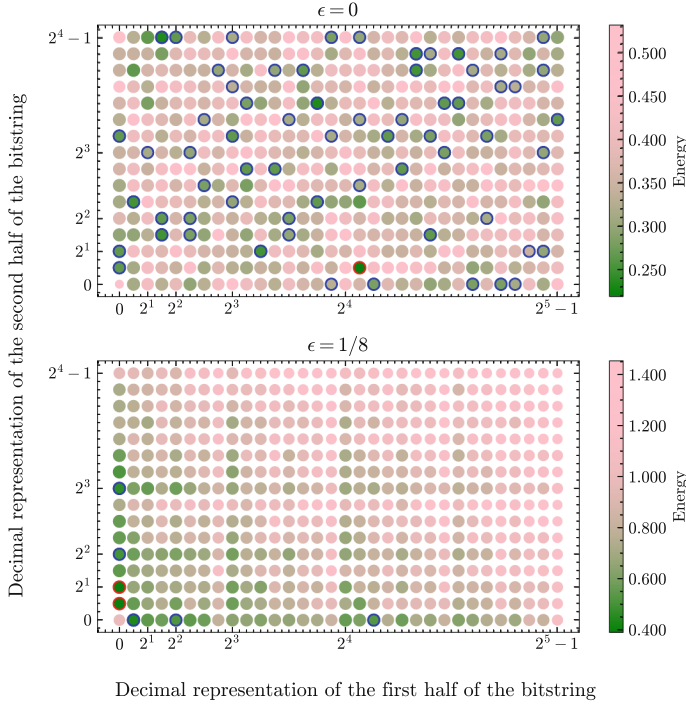


Fig. 1. Hinged bitstring map (HBM) of the 4-glass dataset with 9 features. The plot at the top illustrates the landscape with $\epsilon = 0$, while the plot at the bottom shows the landscape when $\epsilon = 1/8$. Local and global optima are highlighted with a blue and red outline, respectively. (Color figure online)

into two halves: the first half of the bitstring is mapped to the x -axis, while the second half is mapped to the y -axis (rounding up in favor of the x -axis, in case of an odd n). Then, each bitstring $\mathbf{b}$ is colored according to its energy, and the optima are outlined: blue for local optima $\mathbf{L}$ and red for global optima $\mathbf{G}$.

This simple visualization, which we call the *hinged bitstring map* (HBM), comes with a drawback. Two neighboring points in the plot may not necessarily be neighboring bitstrings. Nevertheless, an HBM allows us to build intuition about changes in the landscape when it is subject to regularization. Due to this arrangement, a “grid”-like structure appears. See for example Fig. 2, the HBM for the 5-heart-c dataset for $\epsilon = 1/8$. Lines of good solutions (in green) emerge due to the high regularization factor. For a point (x, y) where $x = 2^k, k \in \mathbb{N}$, its left neighbor $(2^k - 1, y)$ has a much higher penalty: $h_P(2^k) = 1$ and $h_P(2^k - 1) = k$. This produces a vertical line of good solutions compared to their left neighbors. The same applies for horizontal lines when $y = 2^k$.

To compare some of the local optima in more detail, we look at the three lowest-energy optima for each of the selected datasets. Table 2 presents these bitstrings, as well as their energy in two different regularization scenarios. When comparing the three lowest-energy local optima in the 4-glass dataset under

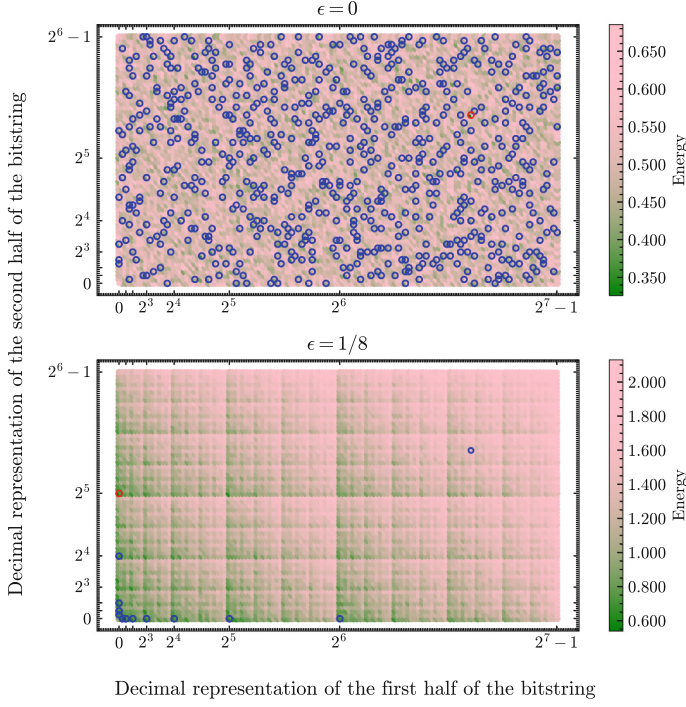


Fig. 2. Hinged bitstring map (HBM) of the 5-heart-c dataset with 13 features. The plot at the top illustrates the landscape with $\epsilon = 0$, while the plot at the bottom shows the landscape when $\epsilon = 1/8$. Local and global optima are highlighted with a blue and red outline, respectively. (Color figure online)

no regularization (the first three rows of Table 2), one can notice some redundant features. This is consistent with previous research on landscape analysis for feature selection [38, 40], where the decision of whether or not to include a redundant feature makes no difference in the model’s accuracy.

However, it is important to also consider that the original local optima (i.e., the local optima of the landscape when $\epsilon = 0$ — $\mathbf{b}_{273}^*$ for 04-glass and $\mathbf{b}_{6571}^*$ for 05-heart-c) are not necessarily optima under another regularization landscape. In fact, notice the *ID* column of Table 2. For both 04-glass and 05-heart-c, the three lowest-energy optima in $\epsilon = 0$ (273, 63 and 235) are not the three lowest-energy optima in $\epsilon = 1/8$, where bitstrings 1, 2, and 16 have the least energy. This notion can be confirmed by looking at Fig. 1, where some of the local optima highlighted in the top panel (with $\epsilon = 0$) are lost when using $\epsilon = 1/8$. To visualize this effect in greater detail, we present plots of the energy of a selection of feature sets in Fig. 3. The selection includes the unregularized, global optima ($\mathbf{b}_{273}^*$ and $\mathbf{b}_{6571}^*$ for 04-glass and 05-heart-c, respectively), as well as some other feature subsets with different number of bits set to 1. See for example $\mathbf{b}_{273}^*$ on the top panel of

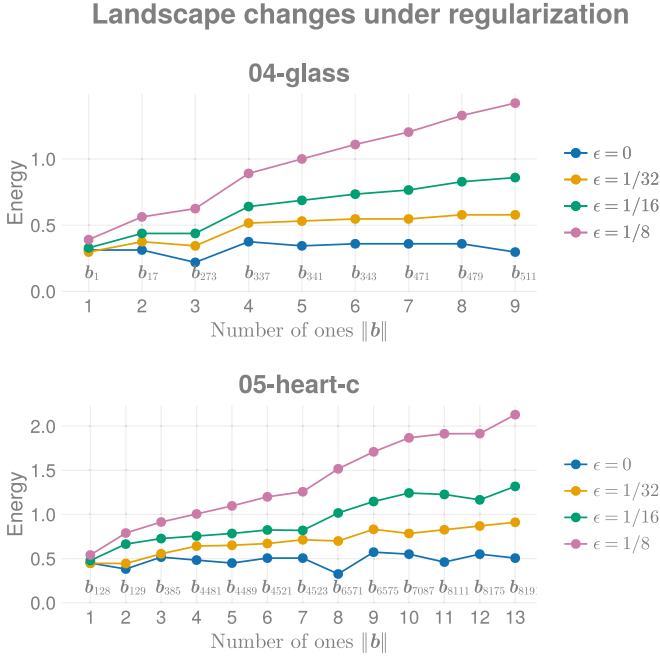


Fig. 3. Energy changes on a fixed set of bitstrings, including 04-glass’ global optimum under $\epsilon = 0$, $\mathbf{b}_{273}^*$, with three features selected (top), and 05-heart-c’s global optimum under $\epsilon = 0$, $\mathbf{b}_{6571}^*$, with eight features selected (bottom).

Fig. 3 (located at $\|\mathbf{b}_{273}^*\|$ on the x -axis), which is no longer an optimum for the other regularization cases $\epsilon \in \{1/32, 1/16, 1/8\}$.

4.3 Experiment 3: Connectedness and Optima Distribution

Goal. The goal of this experiment is to further analyze the 4-glass and 5-heart-c datasets to examine how the spread of optima through the landscape changes under different regularization levels.

Design. For each dataset (i.e., 4-glass and 5-heart-c) and under regularization schemes with $\epsilon \in \{0, 1/8, 1/16, 1/32\}$, we calculate the Hamming distance from all local optima to their closest global optimum in order to look for potential big valley structures [40, 41]. Further, we reconstruct the HBMs (see Figs. 1 and 2) by using LONs. The LONs allow us to see the size of basins of attraction and the links between these basins, things not visualized using the HBMs.

Results and Discussion. Fig. 4 presents a series of hexagonal binned plots of the Hamming distances from all local optima $\mathbf{L}$ to their closest global optimum in $\mathbf{G}$. This plot aggregates the distance counts into bins, where denser regions

Table 2. The tree lowest-energy optima in 4-glass and 5-heart-c datasets, for regularization values $\epsilon = 0$ (top three rows in sub-table) and $\epsilon = 1/8$ (bottom three rows). Redundant or unimportant features are highlighted on each group when there is a tie, i.e., two different feature subsets $\mathbf{b}_i^*$ and $\mathbf{b}_j^*$ have the same energy $h(\mathbf{b}_i^*) = h(\mathbf{b}_j^*)$.

04-glass									
	ID			Bitstring			Energy		Accuracy
	i	x -axis	y -axis	original	x -axis	y -axis	$\epsilon = 0$	$\epsilon = 1/8$	
$\mathbf{b}_{273}^*$	273	17	1	100010001	10001	0001	0.2188	0.5938	0.7813
$\mathbf{b}_{63}^*$	63	3	15	000111111	00011	1111	0.2344	0.9844	0.7656
$\mathbf{b}_{235}^*$	235	14	11	011101011	01110	1011	0.2344	0.9844	0.7656
$\mathbf{b}_1^*$	1	0	1	000000001	00000	0001	0.2656	0.3906	0.7344
$\mathbf{b}_2^*$	2	0	2	000000010	00000	0010	0.2656	0.3906	0.7344
$\mathbf{b}_{16}^*$	16	1	0	000010000	00001	0000	0.3125	0.4375	0.6875
05-heart-c									
$\mathbf{b}_{6571}^*$	6571	102	43	1100110101011	1100110	101011	0.3258	1.3258	0.6742
$\mathbf{b}_{1466}^*$	1466	22	58	0010110111010	0010110	111010	0.3371	1.2121	0.5843
$\mathbf{b}_{2528}^*$	2528	39	32	0100111100000	0100111	100000	0.3371	0.9621	0.5618
$\mathbf{b}_{32}^*$	32	0	32	0000000100000	0000000	100000	0.4157	0.5407	0.5843
$\mathbf{b}_{128}^*$	128	2	0	0000010000000	0000010	000000	0.4382	0.5632	0.5618
$\mathbf{b}_{512}^*$	512	8	0	0001000000000	0001000	000000	0.4494	0.5744	0.5506

(with more counts) have a darker shade. For clarity, regions with no counts are not drawn. Under no regularization, the plots on both datasets illustrate a structure where most of the local optima are concentrated around a single global optimum—a landscape feature often referred to as *the big valley* structure [32, 40, 41]. However, regularization induces changes in the landscape that alter the basins of attraction, and there is a split into separate regions. In other words, the big valley structure disappears for higher ϵ values, here $\epsilon = 1/16$ and $\epsilon = 1/8$. Both of these separate regions are centered on different local optima representing a singleton feature subset. See for example the second and fourth groups of Table 2 (with bitstring IDs $\{1, 2, 16\}$ and $\{32, 128, 512\}$ respectively.) For 04-glass, the number of global optima goes from $|\mathbf{G}| = 1$ with no regularization ($\mathbf{b}_{273}^*$), up to $|\mathbf{G}| = 2$ with regularization $\epsilon = 1/8$ ($\mathbf{b}_1^*$ and $\mathbf{b}_2^*$), both bitstrings 1 and 2 being the center of their own basin in the *split* structure.

The *split* of the landscape can be confirmed by looking at Figs. 5 and 6, which show the resulting LONs for the 4-glass and 5-heart-c datasets respectively. The LONs are plotted with the bidimensional mapping used in the HBMs, in Figs. 1

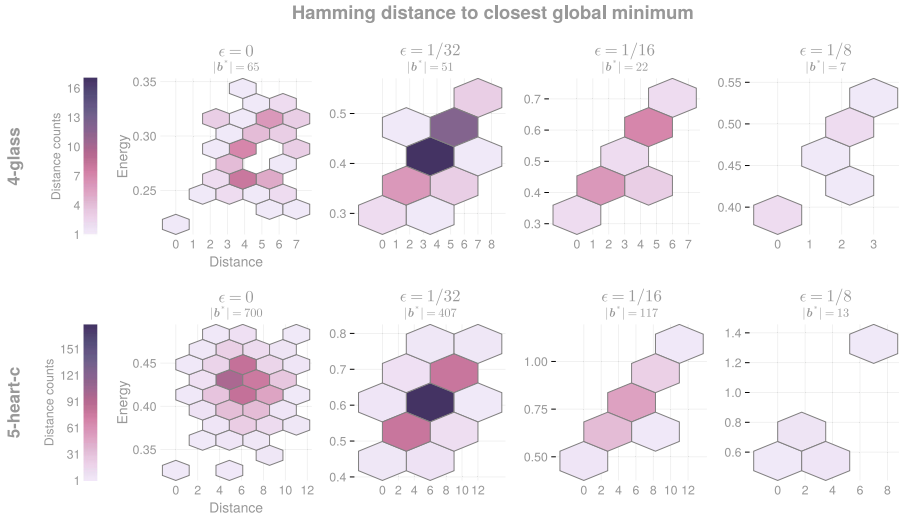


Fig. 4. Hexagonal binned plot of the Hamming distance from all local optima to their closest global optimum. Each bin aggregates distance counts, where a darker shade means more local optima are at that given distance to the global optimum.

and 2. Similar to these HBMs, the color of a vertex corresponds to its energy. In this case, the size of a vertex represents the size of the basin of attraction of optima, and the edges' weights are proportional to the thickness of the drawn lines. To reduce the number of edges drawn and ensure readability (especially in 5-heart-c with $|L| = 700$), edges whose weight is below a certain threshold are discarded. This threshold is set to 30% of the highest weight for the 4-glass dataset and 40% for the 5-heart-c dataset.

In the 04-glass unregularized landscape (c.f. the top left panel of Fig. 5), we can see an optimum that is clearly bigger than the rest, i.e., the center of a big valley. In contrast, the landscape induced by the regularization value of $\epsilon = 1/8$ (c.f. bottom right panel of Fig. 5) shows two *valleys*, around the two global optima near the origin: $\mathbf{b}_{1466}^*$ and $\mathbf{b}_{2528}^*$.

More interestingly, for the 05-heart-c dataset (c.f. Figure 6) the number of optima is considerably higher, so changes in the landscape induced by regularization seem more noticeable. The optima distribution across the landscape goes from an even spread (as seen in the top left panel) to two separate regions with multiple basins of roughly the same size (in the bottom right panel), in line with the *split* shown in the last panel of Fig. 4.

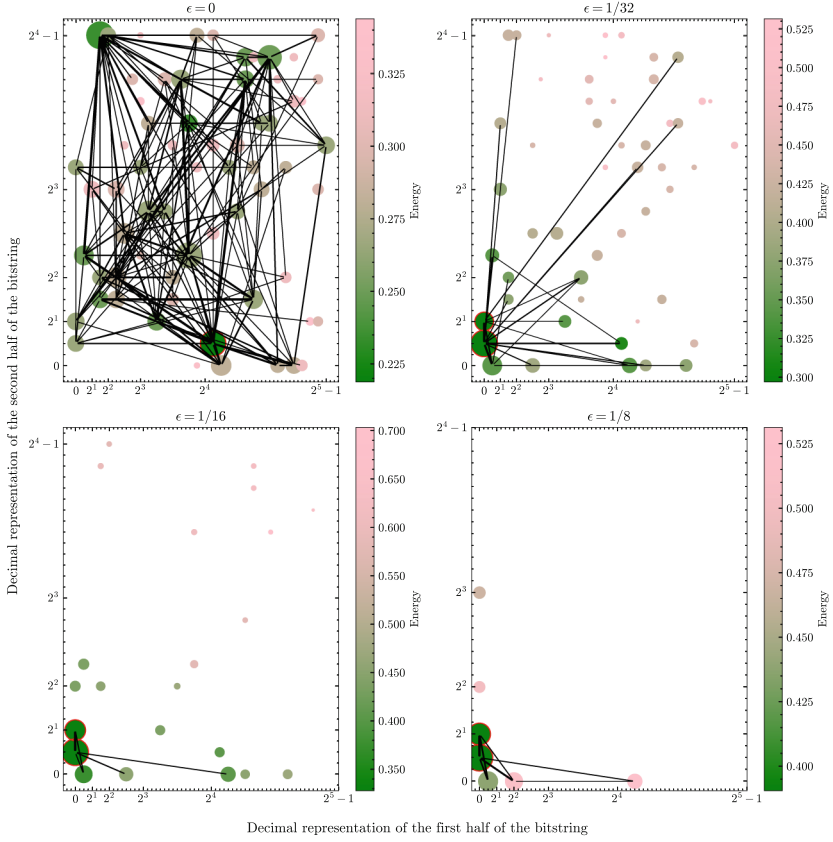


Fig. 5. Partial LONs of the 04-glass dataset (with $n = 9$ features and $|\mathbf{B}^*| = 65$ local optima for $\epsilon = 0$), as subjected to different regularization scenarios. The size of a vertex indicates the size of the basin of attraction, while the thickness of an edge represents its weight. Layout is according to the HBM for 04-glass in Fig. 1.

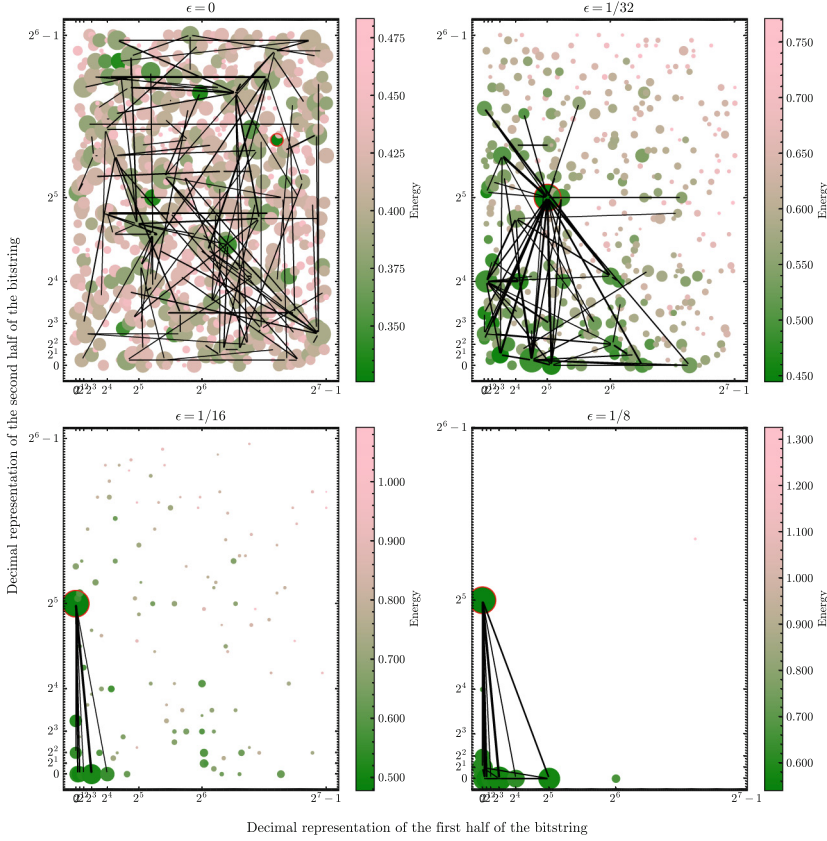


Fig. 6. Partial LONs of the 05-heart-c dataset (with $n = 13$ features and $|\mathcal{B}^*| = 700$ local optima for $\epsilon = 0$), as subjected to different regularization scenarios. Layout is according to the HBM for 05-heart-c in Fig. 2.

5 Conclusion and Future Work

A central component of machine learning (ML) is the processing of features from input data [15]. When it comes to ML in support of high-stakes decision-making, the interpretability of ML models is essential [3, 27, 39, 48]. And feature selection, along with an interpretable ML model like the decision tree, can play an important role in ensuring such interpretability. While a plethora of heuristic methods have been developed and tested for feature selection [7, 13, 18, 19, 22, 26, 34, 62], there has been very little research on the underlying pseudo-Boolean fitness landscape induced by feature selection, with notable exceptions [38]. Previous research indicates that feature selection induces multimodal fitness landscapes [34], but the details of such landscapes have remained largely unknown.

This paper seeks to address this gap in knowledge through an empirical study of the learning of decision trees [4, 46, 47] from ten datasets while varying

the degree of regularization. The modest dataset sizes enable exhaustive enumeration, in which local (including global) optima can easily be identified. We consider the impact of regularization on the number and structure of optima in the fitness landscapes. We conclude that there is a high degree of multimodality when there is no regularization and that the degree of multimodality generally drops off with increasing regularization.

This work improves the understanding of the multimodal nature of the feature selection problem. We hope that this empirical study will inform future research on improved multimodal optimization methods, including evolutionary niching methods, for feature selection and other combinatorial problems. The computed accuracy tables are available online [54].

There are several avenues for extending this research, e.g., carrying out similar analyses on the remainder of the datasets. Another path for future work is to study different types of datasets and ML methods, for example for regression, and explore the impact of the datasets' properties on regularization. Furthermore, combining the analysis with other landscape features (like ruggedness and deception) is a possibility, as is utilizing the newfound knowledge about feature selection fitness landscapes to improve heuristics for feature selection.

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